

Hivemind

Three control-plane attacks

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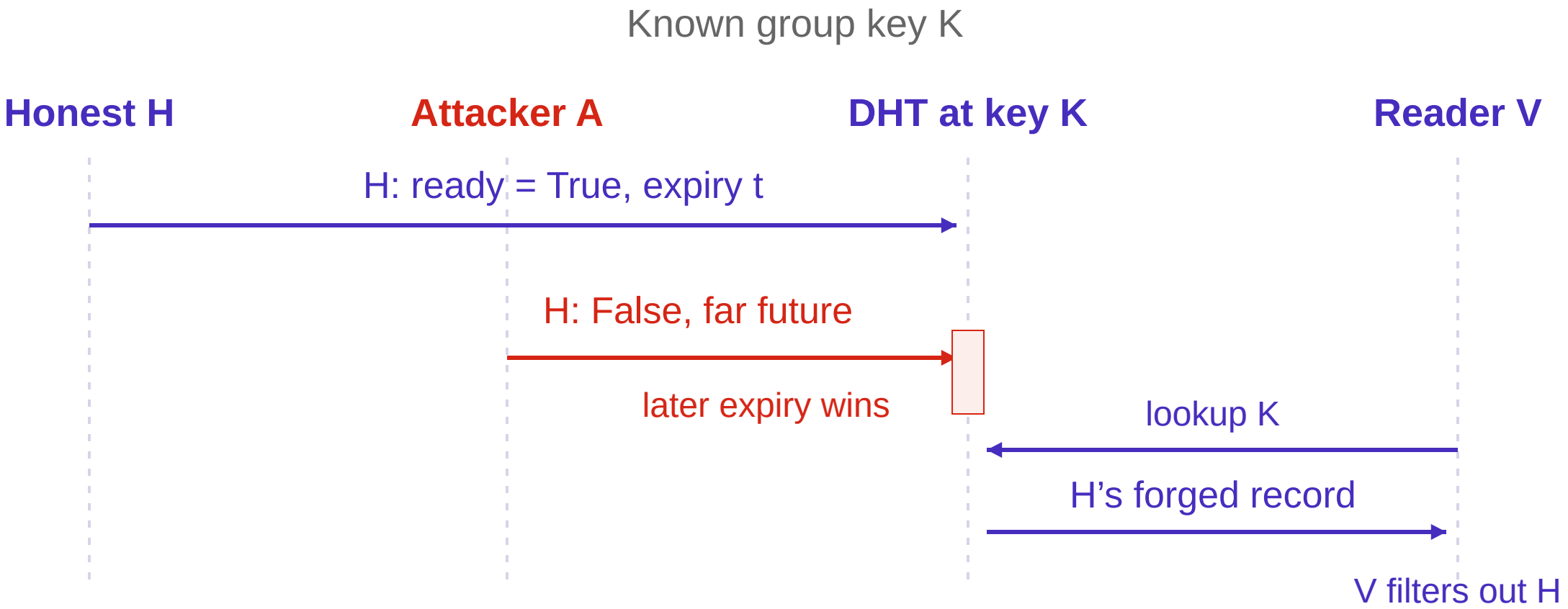
Secura Lab

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Last week → today

Last week's questions	Current status
Can we force leadership without knowing the group key K?	Ongoing
Crawl a live swarm every d seconds and record who has advertised.	No live averaging swarm found. Need a known group key to crawl.
Measure lookup time and the shortest advertised deadline.	Local tests using RIPE Atlas ping RTTs: Sweden, US, Brazil, India, Australia.

1. Forged withdrawal



H stays online, but disappears from group discovery.

H’s refresh expires sooner; the DHT keeps the record that expires later.

Withdrawal: two separate tests

1 Can an attacker hide honest peers?

12 or 40 honest advertisers, plus one attacker.

Check whether forged withdrawals hide peers, even after they advertise again.

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2 Does hiding a peer make the target join the attacker?

Use RIPE Atlas delays in a local swarm.

Compare who the target joins, with and without a forged withdrawal.

Withdrawal: results

Check	Observed result
Targeted peers hidden	39/39 (40 honest peers)
Refreshes that restored the peer	0/30 attempts
Target joined attacker, with withdrawal	0/10 trials
Target joined attacker, without withdrawal	0/10 trials

Group capture was tested locally with delays derived from RIPE Atlas pings.

Exact tables → A1

2. Deadline inflation

Keep H visible, but move its deadline behind the attacker's.

Before

Peer	Advertised deadline
H	now + 5 s
A	now + 6 s

V would contact H

forge H's expiration

After forged write

Peer	Advertised deadline
A	now + 6 s
H	now + 1 hour

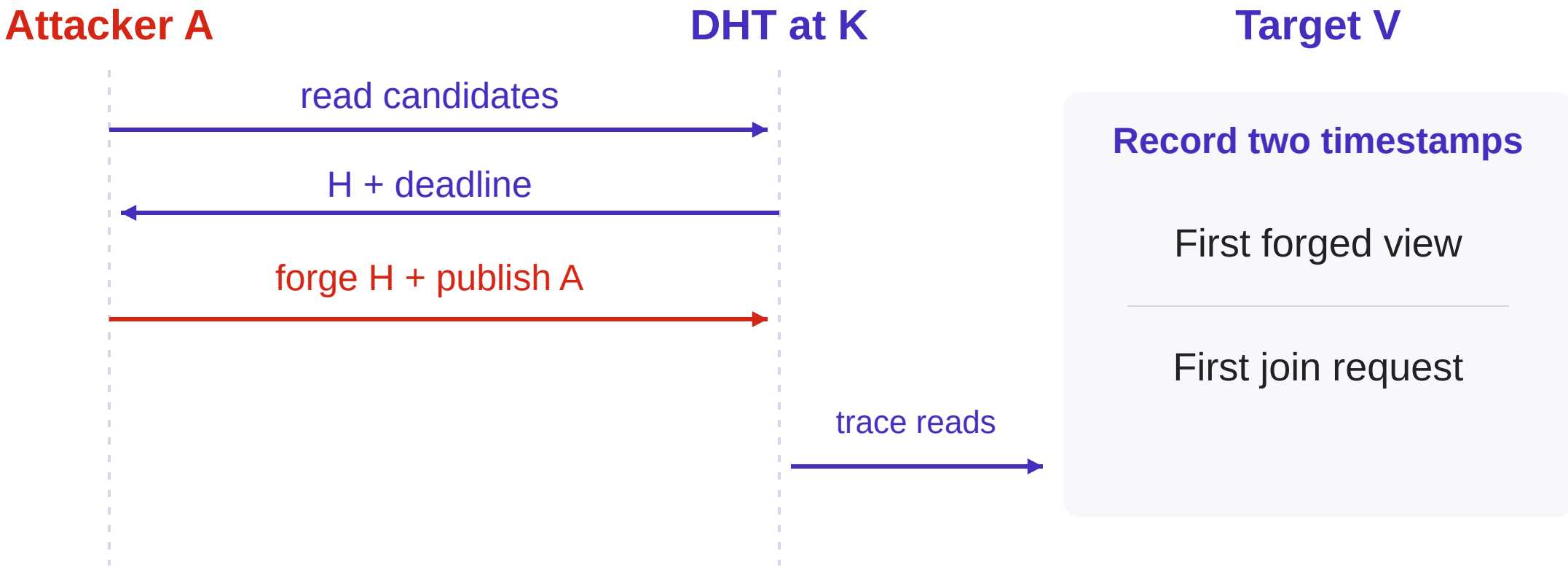
V may contact A

Only if V reads the changed record before choosing whom to join.

Illustrative times · H's ready flag stays True · no crafted DHT identity

Inflation: measure the race to V’s view

A’s own deadline stays later than H’s; only H’s record is forged.



Compare timestamps: forged view before first join request?

28 local DHT nodes · 5 RTT origins · 5–7 s / 11–15 s windows · paired control

Accepted writes did not change V’s join

Measured event	Attack trials
DHT accepted the forged record	**10 / 10**
Forged view recorded before V’s first request	**0 / 10**
V joined the attacker	**0 / 10**

No-forgery control: 0 / 10 targets joined A.

In 7/10 trials: V picked a peer → forged view appeared later

One reactive, single-peer policy; other strategies remain untested.

Exact tables → A2

3. One group, conflicting reducer indices

Precondition: A is already the group leader.



different
ordered rosters

Same group ID
Same four peers

Reducer index	0	1	2	3
A's own view	A	F1	F2	F3
Sent to F1	F1	A	F2	F3
Sent to F2	F2	F1	A	F3
Sent to F3	F3	F1	F2	A

Each recipient claims A's reducer slot (index 0 in this example).

Equal-sized slices can be mixed; followers do not cross-check one common roster.

Roster test: change only the ordering

1 Same inputs

A	0	1	2	4
F1	0	1	2	4
F2	0	1	2	4
F3	0	1	2	4

Four real local averagers

2 Two conditions

One common roster

5 paired trials

Different roster per peer

3 Check outputs

Expected average

0	1	2	4

Did it complete?

Is the tensor correct?

Were failures recorded?

Real DHT + join streams + stock all-reduce · leadership arranged in advance

Wrong tensors, successful steps

One recorded trial

Correct average

0	1	2	4

One attacked peer’s output

1.75	0.75	1.5	3.0

Across five paired trials

Corrupted trials

Control	**0 / 5**
Attack	**5 / 5**

20/20 attacked outputs wrong

All 40 steps completed; tested failure indicators stayed clear.

Tensor corruption measured locally; downstream training damage was not tested.

Exact tables → A3

A1. Withdrawal discovery tests

One ordinary random-ID writer · counts below matched in both repeats

Honest peers	Targets / run	Hidden / run	Others visible	Repeats
12	1	1	11 / 11	2
12	3	3	9 / 9	2
12	6	6	6 / 6	2
12	11	11	1 / 1	2
40	1	1	39 / 39	2
40	20	20	20 / 20	2
40	39	39	1 / 1	2

Separate refresh test: 0/30 attempts restored H in each swarm size.

Source: step8_withdrawal_phase2.csv and step8_withdrawal_phase2_n40.csv

Exact tables → A2

A2. Forged-record trials

Targets that joined A / trials · one seed per region and deadline window

RTT origin	Window	Control	Withdrawal	Inflation
Sweden	5-7 s	0 / 1	0 / 1	0 / 1
United States	5-7 s	0 / 1	0 / 1	0 / 1
Brazil	5-7 s	0 / 1	0 / 1	0 / 1
India	5-7 s	0 / 1	0 / 1	0 / 1
Australia	5-7 s	0 / 1	0 / 1	0 / 1
Sweden	11-15 s	0 / 1	0 / 1	0 / 1
United States	11-15 s	0 / 1	0 / 1	0 / 1
Brazil	11-15 s	0 / 1	0 / 1	0 / 1
India	11-15 s	0 / 1	0 / 1	0 / 1
Australia	11-15 s	0 / 1	0 / 1	0 / 1

Forgery accepted in 20/20 attack trials (29/29 writes); all 90 honest steps completed.

28-node local RTT emulation · source: step20 sweep_20261002_a analysis + event CSVs

A3. Reducer-index trials

Four peers · five paired local trials · identical input [0, 1, 2, 4]

Trial	Control correct	Attack correct	Max. abs. error	Steps
1	4 / 4	0 / 4	3.00	8 / 8
2	4 / 4	0 / 4	3.00	8 / 8
3	4 / 4	0 / 4	3.00	8 / 8
4	4 / 4	0 / 4	2.25	8 / 8
5	4 / 4	0 / 4	3.00	8 / 8

No bans, fallback elements, RPC/all-reduce errors, timeouts, or hard cancellations.

Source: step17_roster_live_rounds_confirm_20260928_confirm_t5_s1717.csv